go-On — Notification Count & Empty Inbox Synchronization

Comprehensive Root Cause Analysis, Fix Implementation, and Verification Report

System: go-On AI Face Attendance	**Target OS:** Android (S24 SM-S921E) & Web	**Status:** RESOLVED & TESTED
Scope: Notifications History UI & Service	**Date:** October 20, 2026	**Analysis:** 0 Issues (Clean)

1. Executive Summary

A critical defect was identified in the Notifications page where the AppBar badge and filter chips displayed unread counts (e.g. **Notifications 3**, **Unread 3**) while the body rendered the empty state: *'No notifications yet. Class updates and announcements will appear here.'* An end-to-end investigation from database models up to UI widget trees uncovered six interrelated root causes, including asynchronous unread polling race conditions, initial post-frame empty state flashes, destructive filter re-fetches, and unsanitized service states on login transitions. All causes have been addressed and verified across automated test suites.

2. Root Cause Analysis Matrix

ID	Subsystem	Root Cause Description	Observed Behavioral Impact	Engineering Fix Applied
RC-1	UI Layer	All chip and Unread chip read from different sources (list length vs counter variable).	All showed 0 while Unread and AppBar showed 3.	Both counts unified to derive strictly from active in-memory notifications.
RC-2	UI Filter	Tapping Unread chip called API with unreadOnly=true, overwriting full list.	All count got corrupted to match unread count; list had to be re-downloaded.	Local in-memory filtering applied. Destructive re-fetch removed.
RC-3	UI Lifecycle	Initial fetch was queued in addPostFrameCallback. Frame 1 had empty items and isLoading=false.	Screen briefly flashed 'No notifications yet' while showing 3 in badges.	Added _hasInitialLoaded lifecycle guard and spinner for initial load.
RC-4	Service Layer	fetchNotifications emitted notifyListeners before reconciling unread count.	Temporary race condition between items and counter.	Immediate activeUnread calculation before notifyListeners.
RC-5	Auth Service	clearUserData was only called on logout, never on login or session restore.	Previous user notification count leaked into next logged-in account.	clearUserData invoked in restoreSession, loginStudent, and loginFaculty.
RC-6	Backend ORM	mark_all_notifications_as_read lacked synchronize_session='fetch'.	Stale cached counts in concurrent sessions.	Added explicit synchronize_session='fetch' to bulk update.

3. Verification and Test Results

Test Suite	Test Scenario / Verification Objective	Result
notification_count_and_sync_test.dart	NotificationService clearUserData resets counts, notifications, and state	**PASSED**
notification_count_and_sync_test.dart	Empty inbox renders zero counts, hidden header badge, and empty state message	**PASSED**
notification_count_and_sync_test.dart	Populated inbox with 3 unread displays matching header badge and filter counts	**PASSED**
notification_count_and_sync_test.dart	Tapping Unread filter tab filters list in-place without destroying All count	**PASSED**
notification_count_and_sync_test.dart	Unread tab with 0 unread displays correct empty state message without corrupting All tab	**PASSED**
faculty_notification_bell_and_delivery_test.dart	Active bell, zero unread hiding, single/multi-digit badge verification (6 tests)	**PASSED (6/6)**
notification_bulk_delete_test.dart	Bulk deletion confirmation, unpinned deletion, pinned preservation (7 tests)	**PASSED (7/7)**
notification_delete_pin_test.dart	Swipe to pin, swipe to delete, system tray deduplication (7 tests)	**PASSED (7/7)**
Backend Pytest	test_notification_bulk_delete.py, test_notification_delete_pin_audit.py, test_phase11_notifications.py	**PASSED (22/22)**
Static Code Analysis	flutter analyze (Entire Mobile Codebase)	**0 ISSUES**

4. Key Architectural Guarantees

• **Zero Count Discrepancies:** The header badge, filter tabs, and inbox list now share identical mathematical derivation.
• **Pure Local Filtering:** Switching tabs does not trigger unnecessary network roundtrips or wipe out in-memory data.
• **Deterministic Empty State:** 'No notifications yet' is only shown when the user truly has zero notifications, and unread empty state specifically states 'All your notifications have been marked as read.'
• **Clean Session Transitions:** Account switching completely purges previous user cache and polling queues.